

D312

Anatomy and Physiology I with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An adult line cook, age 32, is burned over the entire right arm and the entire front of the trunk in a kitchen fire. The fire started at 2 a.m. after closing. Using the rule of nines, what percentage of body surface is burned?

- A. 27%
- B. 32%
- C. 18%
- D. 45%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An adult has burns covering 18% of body surface, all in a single region. Using the rule of nines, which region could it be?

- A. One entire leg
- B. One entire arm
- C. Both legs
- D. The front of one leg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A lab panel shows a woman's blood calcium is below normal. She runs three miles every morning and drinks two cups of coffee a day. Which response will the body use to raise blood calcium?

- A. The pancreas releases insulin.
- B. The thyroid releases calcitonin.
- C. The parathyroid glands release PTH, which increases osteoclast activity.
- D. Osteoblasts increase bone building.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Blood calcium rises above normal. Which change helps bring it back down?

- A.** Aldosterone release from the adrenal cortex.
- B.** Calcitonin release from the thyroid, favoring calcium deposit into bone.
- C.** Increased osteoclast activity.
- D.** More PTH release.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

On a hot afternoon a hiker's body temperature rises. She is wearing a blue backpack and has walked four miles. Her sweat glands increase output until her temperature returns toward 37 °C. What type of control is this?

- A.** Positive feedback, because sweat output increased
- B.** Negative feedback
- C.** Positive feedback, like contractions during childbirth
- D.** No feedback; it is caused by the four-mile walk

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A motor neuron releases acetylcholine at a skeletal muscle fiber, and calcium is released from the sarcoplasmic reticulum. The athlete had eaten a banana an hour earlier. What does calcium do next to allow cross-bridges to form?

- A.** It binds troponin, which shifts tropomyosin off the binding sites on actin.
- B.** It supplies potassium from the banana to the sarcomere.
- C.** It binds myosin to detach it from actin.
- D.** It binds acetylcholine receptors on the fiber.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A neuron in a lab model reaches threshold, and voltage-gated channels open. The model was built by a team of four students. Which ion movement causes the rapid depolarization?

- A.** Stronger stimuli let more sodium in and make a bigger action potential.
- B.** Sodium ions flowing out of the cell.
- C.** Sodium ions flowing into the cell.
- D.** Potassium ions flowing out of the cell.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A hiker in a national park suddenly sees a bear 30 feet away. He had just finished lunch and was checking his map. Which set of changes will his autonomic nervous system produce?

- A. Constricted bronchioles to conserve air.
- B. Increased digestion because he just ate.
- C. Faster heart rate, dilated pupils, and slowed digestion.
- D. Slower heart rate and increased digestion.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A person is relaxing on a couch after a large dinner. Which effect is expected from the dominant autonomic division?

- A. Dilated pupils and faster heart rate.
- B. Release of glucose from the liver for energy.
- C. No autonomic activity at all while resting.
- D. Increased stomach and intestinal activity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A soccer player has a bruise on the shin, between the knee and the ankle. She plays midfield and wears size 8 cleats. Relative to the knee, where is the bruise?

- A. Inferior only; distal is not used on the leg
- B. Lateral
- C. Proximal
- D. Distal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Using directional terms, how is the elbow described relative to the wrist?

- A. Distal
- B. Superficial
- C. Medial
- D. Proximal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Cortisol, a steroid hormone made from cholesterol in the adrenal cortex, is released during stress. Levels usually peak in the early morning. How does cortisol act on its target cells?

- A.** It binds a surface receptor and activates cAMP as a second messenger.
- B.** It crosses the cell membrane and binds an intracellular receptor that changes gene expression.
- C.** It cannot enter cells because it is too large.
- D.** It only acts in the morning when levels peak.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

An adult line cook, ~~age 32~~, is burned over the entire right arm and the entire front of the trunk in a kitchen fire. ~~The fire started at 2 a.m. after closing.~~ Using the rule of nines, what percentage of body surface is burned?

Target word: total % burned **Decides:** Arm 9% + front trunk 18%.

A	Correct	9% + 18% = 27%.
B	Wrong stem number	The patient's age, not a burn area.
C	Dropped element	Counts only the trunk and leaves out the arm.
D	Adjacent category	Counts front and back of the trunk (36%) plus the arm.

2. Answer A Concept 1 • transfer

An adult has burns covering 18% of body surface, all in a single region. Using the rule of nines, which region could it be?

Target word: reverse: percentage to region **Decides:** Regions worth 18% alone: one entire leg, front trunk, back trunk.

A	Correct	Each whole leg is 18%.
B	Confusion set	Each arm is 9%.
C	Adjacent category	Both legs total 36%.
D	Dropped element	Half a leg is 9%.

3. Answer C Concept 2

A lab panel shows a woman's blood calcium is below normal. ~~She runs three miles every morning and drinks two cups of coffee a day.~~ Which response will the body use to raise blood calcium?

Target word: hormone response to low calcium **Decides:** Low calcium triggers PTH.

A	Confusion set	Insulin regulates glucose, not calcium.
B	Inverted operation	Calcitonin lowers blood calcium.
C	Correct	Osteoclasts break down bone matrix and release calcium.
D	Adjacent category	Osteoblasts deposit calcium into bone, lowering blood levels.

4. Answer B Concept 2 • transfer

Blood calcium rises above normal. Which change helps bring it back down?

Target word: reverse: response to high calcium **Decides:** High calcium triggers calcitonin.

A	Confusion set	Aldosterone regulates sodium and potassium.
B	Correct	Calcitonin lowers blood calcium.
C	Adjacent category	Osteoclasts release calcium into blood.
D	Inverted operation	PTH raises blood calcium.

5. Answer B Concept 3

On a hot afternoon a hiker's body temperature rises. ~~She is wearing a blue backpack and has walked four miles.~~ Her sweat glands increase output until her temperature returns toward 37 °C. What type of control is this?

Target word: feedback type **Decides:** The response reverses the change.

A	Inverted operation	Feedback type depends on the effect on the variable, not whether output rises.
B	Correct	Sweating counters the rise and restores the set point.
C	Confusion set	Childbirth amplifies change; sweating reverses it.
D	Camouflage	The walk explains the heat, not the control mechanism.

6. Answer A Concept 4

A motor neuron releases acetylcholine at a skeletal muscle fiber, and calcium is released from the sarcoplasmic reticulum. ~~The athlete had eaten a banana an hour earlier.~~ What does calcium do next to allow cross-bridges to form?

Target word: role of calcium **Decides:** Calcium binds troponin.

A	Correct	This exposes actin for myosin heads.
B	Camouflage	Diet does not decide this step.
C	Inverted operation	Detachment requires ATP, not calcium.
D	Confusion set	ACh, not calcium, binds those receptors.

7. Answer C Concept 5

A neuron in a lab model reaches threshold, and voltage-gated channels open. ~~The model was built by a team of four students.~~ Which ion movement causes the rapid depolarization?

Target word: depolarization ion **Decides:** Na⁺ influx.

A	Adjacent category	Action potentials are all-or-none; intensity is coded by frequency.
B	Inverted operation	Right ion, wrong direction.
C	Correct	Na ⁺ influx makes the inside more positive.
D	Confusion set	K ⁺ efflux causes repolarization.

8. Answer C Concept 6

A hiker in a national park suddenly sees a bear 30 feet away. ~~He had just finished lunch and was checking his map.~~ Which set of changes will his autonomic nervous system produce?

Target word: sympathetic response **Decides:** A threat triggers fight or flight.

A	Adjacent category	Sympathetic activation dilates bronchioles.
B	Camouflage	The threat overrides the after-meal state.
C	Correct	Classic sympathetic effects.
D	Inverted operation	Those are parasympathetic effects.

9. Answer D Concept 6 • transfer

A person is relaxing on a couch after a large dinner. Which effect is expected from the dominant autonomic division?

Target word: parasympathetic effect **Decides:** Rest-and-digest state.

A	Inverted operation	Those are sympathetic effects.
B	Confusion set	That is a sympathetic, energy-mobilizing effect.
C	Absolute language	The parasympathetic division is active at rest.
D	Correct	Parasympathetic activity promotes digestion.

10. Answer D Concept 7

A soccer player has a bruise on the shin, between the knee and the ankle. ~~She plays midfield and wears size 8 cleats.~~ Relative to the knee, where is the bruise?

Target word: direction along a limb **Decides:** Farther from the trunk attachment than the knee.

A	Absolute language	Proximal and distal are the preferred terms along limbs.
B	Adjacent category	Lateral refers to distance from the midline, not along the limb.
C	Inverted operation	Proximal means closer to the attachment point.
D	Correct	The shin is farther from the hip than the knee is.

11. Answer D Concept 7 • transfer

Using directional terms, how is the elbow described relative to the wrist?

Target word: reverse the reference point **Decides:** Elbow is closer to the shoulder.

A	Inverted operation	The wrist is distal to the elbow, not the reverse.
B	Adjacent category	Superficial describes depth, not position along a limb.
C	Confusion set	Medial relates to the body's midline.
D	Correct	The elbow is nearer the limb's attachment than the wrist.

12. Answer B Concept 8

Cortisol, a steroid hormone made from cholesterol in the adrenal cortex, is released during stress. ~~Levels usually peak in the early morning.~~ How does cortisol act on its target cells?

Target word: mechanism of a steroid **Decides:** Steroids are lipid-soluble.

A	Confusion set	That is how peptide hormones act.
B	Correct	Lipid-soluble hormones act inside the cell.
C	Inverted operation	Steroids pass readily through the lipid membrane.
D	Camouflage	Timing of peak levels does not describe the mechanism.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Rule of nines (burn area)	In an adult: head and neck 9%, each arm 9%, front of trunk 18%, back of trunk 18%, each leg 18%, genital area 1%.	Giving an arm 18% or counting the whole trunk as 18%.	Burned body regions and a request for total body surface area.
2	Calcium balance and bone	Low blood calcium triggers parathyroid hormone, which activates osteoclasts to release calcium from bone; high blood calcium triggers calcitonin, which favors deposit into bone.	Swapping PTH and calcitonin, or osteoblasts and osteoclasts.	Blood calcium above or below normal.
3	Negative vs. positive feedback	Negative feedback reverses a change to restore a set point (most homeostasis); positive feedback amplifies a change until an endpoint (e.g., childbirth, clotting).	Calling any response that increases something positive feedback.	Whether the response pushes the variable back toward normal or further away.
4	Muscle contraction (sliding filament)	Calcium binds troponin, moving tropomyosin off actin's binding sites so myosin can attach; ATP is needed for myosin to detach and re-cock.	Thinking ATP causes binding, or that ACh binds troponin.	The step that exposes binding sites or allows release.
5	Action potential	Na ⁺ rushing into the neuron depolarizes it; K ⁺ leaving repolarizes it; the action potential is all-or-none once threshold is reached.	Thinking stronger stimuli make bigger action potentials.	Which ion movement causes which phase.
6	Sympathetic vs. parasympathetic	Sympathetic = fight or flight (heart rate up, pupils dilate, bronchi dilate, digestion slows); parasympathetic = rest and digest (the reverse).	Assigning digestion to the sympathetic side.	A threat or a calm, after-meal state.
7	Directional terms	Proximal = closer to the limb's attachment to the trunk; distal = farther; medial = toward the midline; lateral = away from it.	Using superior/inferior or medial/lateral for limb position along the limb.	Two points on the same limb.
8	Steroid vs. peptide hormones	Steroid hormones are lipid-soluble, cross the membrane, and bind intracellular receptors to change gene expression; peptide hormones bind membrane receptors and act through second messengers.	Saying steroids need second messengers.	A hormone's chemical class and where its receptor is.

Course scope (public description): This is Anatomy and Physiology I, a six-section, 4 CU course that enables students to develop an understanding of the relationships between the structures and function of the integumentary, skeletal, muscular, nervous and endocrine systems in the human body. This course will involve laboratory activities, simulated dissections, textbook material, models, and diagrams. Because the course is self-paced, you may move through the material as quickly or as slowly as you need to, with the goal of demonstrating proficiency in the four competencies covered in the final assessment. If you have no prior knowledge of this

material, you can expect to spend 40–60 hours on the course content.